

The Rate Function Carries the Mirror

Large Deviations and the Legendre-Fenchel Dual of the Tilt, from $P(+W)/P(-W) = e^W$

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L.O.R.E. Framework (Law of Repulsive Emanation)

Abstract

Ch.77 measured the detailed mirror $M(w) = \ln P(w) - \ln P(-w) = w$, and Ch.78 showed the same equality makes the entire cumulant ladder obey $k_{-n} = (-1)^n k_n$ under the $e^{(-W)}$ tilt. This chapter asks what the mirror looks like at the level where the tail actually lives: the large-deviation rate function $I(a)$, the Legendre-Fenchel conjugate of the moment-generating function $\phi(t) = \langle e^{(-tW)} \rangle = \phi(1-t)$. For the Gaussian (Onsager-Machlup 1953, Mazonka-Jarzynski 1999 linearity) work distribution, $I(a) = (a-\mu)^2/(2\sigma^2)$ is a quadratic, EVEN about the mean - $I(a) = I(2\mu - a)$ - and the tilt shifts its mode by exactly one variance, $a^* = \mu - \sigma^2 = k_{-1}$ of Ch.78. The Ch.78 empty $n \geq 3$ rows are exactly the emptiness of a quadratic: its whole tail is carried in two numbers. A measured mirror RESIDUAL $R(a) = I(a) - I(2\mu - a)$ is exactly 0 for the parabola and rises to +3.06 in the control's far tail (skew +1.30, $q = 0.90$) - the tail deposit of the TUR shape - deepening to +4.01 engaged and +5.14 harvest. The coin moves the whole tail at once.

1. From the Ledger to the Curve

Ch.78 read the Taylor coefficients of $F = \ln \phi$ at one point. The large-deviation principle reads the whole function: for a long succession of n draws, $P(W \sim a n) \sim \exp(-n I(a))$ with

$$I(a) = \sup_t [a t - \ln \phi(t)] \quad (\text{Gartner-Ellis / Cramer})$$

The exchange symmetry $\phi(t) = \phi(1-t)$ is a statement about the integrand; its conjugate is a statement about how EXPENSIVE different averaged work values are. Because the measure is normalized on its support, the DUAL legendre-Fenchel pair carries no new parameter - it is the same mirror, one level up, and the same 0/0 regimes appear as derivatives that vanish together.

2. The Quadratic That Is a Whole Book

For Gaussian work, $F(t) = \mu t + \sigma^2 t^2/2$, so the conjugate is exactly

$$I(a) = (a - \mu)^2 / (2 \sigma^2) \quad (\text{Onsager-Machlup 1953})$$

$$\text{tilted mode: } dI/da = 0 \rightarrow a^* = \mu - \sigma^2 = k_{-1} \quad (\text{Ch.78})$$

$$\text{mirror: } I(a) = I(2\mu - a) \text{ exactly; } R(a) = 0 \text{ to } 3e-14 \text{ measured}$$

The tilt subtracts exactly one variance from the most-likely work. On the drag parabola ($\mu = +5.98$, $\sigma^2 = 12.04$, $q = 0.9926$): the mode is pushed to $-6.07 = \mu - \sigma^2$, EXACTLY the k_{-1} that Ch.78's ladder reported. And because $\sigma^2 = 2\mu$ (VDT of Ch.72), the tilted mode sits at the mirror image $a^* = -\mu$ of the untilted $a^* = +\mu$ - the detailed mirror at the level of the curve. The 0/0 of Ch.78 (every $n \geq 3$ cumulant empty in both columns) is simply the statement that all derivatives of a quadratic beyond two vanish: a quadratic rate function carries its ENTIRE tail in two numbers, and there is no third piece of information to mirror.

3. The Mirror Residual on the Control

The no-feedback round trip ($q = 0.90$, skew +1.30, kurt 7.0) has a NON-Gaussian rate function. Define the mirror residual

$$R(a) = I(a) - I(2\mu - a) \quad (\text{Gaussian: } R = 0, \text{ entire tail})$$

$$\text{measured control: } R(a) = +3.06 \text{ at } a = 2.57 \text{ (far tail)}$$

R is exactly zero where the PDF mirror holds at rate level (the quadratic) and POSITIVE in the far tail of the control - the skew's deposit, the same shape that Ch.76 priced as $q = 0.894$ and Ch.77 showed lay off the mirror. The detailed theorem fixed every RATIO $P(w)/P(-w)$; it says nothing about the SHAPE, and the shape shows up here as the tail asymmetry that a quadratic cannot carry.

4. The Coin Is in the Far Tail of Both

Feedback leaves the residual cumulative:

row	μ	σ^2	q	far-tail $R(a)$
control	0.113	0.251	0.90	+3.06
engaged	0.35/2	0.045	0.259	+4.01
harvest	0.05/16	-0.042	0.193	+5.14

One coin, shown twice: the mode sits further from 2μ (the mean itself moves toward the bit) and the far-tail residual deepens with each stronger coin. The rate function is the integrated ledger - Ch.76 priced knowing in variance, Ch.78 in cumulants, and here the whole tail of $I(a)$ leans by the same one entry.

Main Results

THEOREM

The Rate Function Carries the Mirror: on the $\Delta F = 0$ round trip, the Gaussian rate function $I(a) = (a-\mu)^2/2\sigma^2$ is even about the mean ($R(a) = 0$ to $3e-14$), its tilt shifts the mode by exactly $-\sigma^2 = k_{-1}$ of Ch.78, and every higher cumulant's emptiness is the quadratic's empty derivatives - the whole tail in two numbers.

COROLLARY

The Coin Is in the Far Tail: the mirror residual $R(a) = I(a) - I(2\mu - a)$ is 0 for the Gaussian, +3.06 on the $q = 0.90$ control (the skew's tail deposit), +4.01 engaged, +5.14 harvest - the same one entry that priced knowing in variance and cumulant now moves the entire rate function.

COROLLARY

One Ledger, Three Books: $M(w) = w$ (Ch.77, PDF), $k_{-n} = (-1)^n k_n$ (Ch.78, cumulants), $I(a) = I(2\mu - a)$ (this chapter, tail) are the same equality $P(+W)/P(-W) = e^W$ read at point, Taylor-coefficient, and whole-curve level; the 0/0 resolves the same way at each.

Conclusion

Where Ch.78 counted cumulants, Ch.79 reads the curve. The Gaussian rate function is a perfect mirror - even about the mean, tilted by one variance - and its emptiness above two numbers is the Ch.78 0/0 dressed as a whole distribution of prices. The non-Gaussian control leaves a tail residual that a quadratic cannot carry, and the coin, once spent, deepens it from +3.06 to +5.14. Six chapters of one strand - integral FT, Gaussian, TUR, detailed mirror, cumulant ladder, and now the rate function - are one equality read at every resolution, and the demon's coin is a single entry in all of them.

References

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- [5] Cramer, H. (1938) & Gartner-Ellis theorem (level-1 large deviations).
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